

Topic to discuss

- Modified Euler's Method
- Numerical Problem
- Home work Problem

Modified Euler's Method

Modified Euler's Method improves Euler's method by using the average of slopes and repeating correction until values stabilize.

This is also called:

→ Heun's Method

→ RK2 Method (2nd order)

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

Step 1: Predictor (Euler step)

$$y_{n+1}^{(0)} = y_n + h f(x_n, y_n)$$

Step 2: Corrector (Average slope)

$$y_{n+1}^{(1)} = y_n + \frac{h}{2} \left[f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(0)}) \right]$$

$$y_{n+1}^{(2)} = y_n + \frac{h}{2} \left[f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(1)}) \right]$$

Note: Repeat until values stop changing
→ we don't stop after 1 correction

Numerical Problem:

Q: Solve by Euler's modified method the following differential equation for $x=0.02$ by taking step length $h=0.01$ (correct to 4 decimal places)

$$y' = x^2 + y, \quad y(0) = 1$$

Solution:

$$\text{Let } f(x, y) = x^2 + y$$

$$x_0 = 0$$

$$y_0 = 1$$

$$h = 0.01$$

We have to find y at $x=0.02$

$$x_1 = x_0 + h = 0 + 0.01 = 0.01$$

$$x_2 = x_1 + h = 0.01 + 0.01 = 0.02$$

So, first we have to find y_1 and y_2 for $y(0.02)$

Step-1 : finding y_1

finding $y(0.01)$, $x=0$

$$x_0 = 0 , y_0 = 1 , h = 0.01$$

$$\text{and } f(x_0, y_0) = 0^2 + 1 = 1$$

$$\Rightarrow y_1^{(0)} = y_0 + h f(x_0, y_0)$$

$$= 1 + 0.01 \times 1$$

$$\therefore y_1^{(0)} = 1.01$$

first approximation: $n=0$

$$y_{n+1}^{(1)} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(0)})]$$

$$\Rightarrow y_1^{(1)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(0)})]$$

$$= 1 + \frac{0.01}{2} [1 + (x_1^2 + y_1^{(0)})]$$

$$= 1 + \frac{0.01}{2} [1 + 0.01^2 + 1.01]$$

$$= 1.0100505$$

$$\text{So, } y_1^{(1)} = 1.0100505$$

Second Approximation :

$$y_1^{(2)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

$$= 1 + \frac{0.01}{2} [1 + 0.01^2 + 1.0100505]$$

$$= 1.01005075$$

$$\therefore y_1^{(1)} = y_1^{(2)} = 1.010050$$

$\therefore$ value of y_1 in 1st approximation & 2nd approximation is same upto 4 decimal places,

Step 2: finding y_2 , $y(0.02)$

Now, $x_1 = 0.01$ $y_1 = 1.01005075$

$$f(x_1, y_1) = x_1^2 + y_1 = 0.01^2 + 1.01005075 = 1.01015075$$

$$\Rightarrow y_2^{(0)} = y_1 + h f(x_1, y_1) \\ = 1.01005075 + 0.01 (1.01015075)$$

$$y_2^{(0)} = 1.020152258$$

first approximation: $n=1$

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(0)})]$$

$$y_2^{(1)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(0)})]$$

$$= 1.01005075 + \frac{0.01}{2} [1.01015075 + x_2^2 + y_2^{(0)}]$$

$$= 1.01005075 + \frac{0.01}{2} [1.01015075 + 0.02^2 + 1.020152258]$$

$$y_2^{(1)} = 1.020204265$$

Second Approximation:

$$y_2^{(2)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(1)})]$$

$$= 1.01005075 + \frac{0.01}{2} \left[1.01015075 + 0.02^2 + 1.020204265 \right]$$

$$y_2^{(2)} = 1.020204525$$

$$\therefore y_2^{(1)} = y_2^{(2)} = 1.020204$$

Value of y_2 at 1st & 2nd approximation is same upto 4 decimal places.

$$\therefore y_2 = y(0.02) = 1.020204 \underline{\underline{Ans.}}$$

Homework Problem

Q:- Given $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2}$, $y(1) = 1$

Evaluate $y(1.2)$ by modified Euler's method
correct up to 4 decimal places.

Solution: We have,

$$\frac{dy}{dx} = y' = f(x, y)$$

Given, $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2}$; $x_0 = 1$; $y_0 = 1$

Let, $h = 0.1$

First Convert Equation :

$$\frac{dy}{dx} = \frac{1}{x^2} - \frac{y}{x}$$

$$\text{So, } f(x, y) = \frac{1}{x^2} - \frac{y}{x}$$

Step 1 : finding $y(1.1)$

$$x_1 = x_0 + h = 1 + 0.1 = 1.1$$

$$\text{and } f(x_0, y_0) = \frac{1}{x_0^2} - \frac{y_0}{x_0} = \frac{1}{1} - \frac{1}{1} = 0$$

$$\Rightarrow y_1^{(0)} = y_0 + h f(x_0, y_0)$$

$$y_1^{(0)} = 1 + 0.1 \times 0$$
$$y_1^{(0)} = 1$$

First approximation .

$$y_1^{(1)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(0)})]$$

$$= 1 + \frac{0.1}{2} \left[0 + \frac{1}{x_1^2} - \frac{y_1^{(0)}}{x_1} \right]$$

$$= 1 + \frac{0.1}{2} \left[\frac{1}{1.1^2} - \frac{1}{1.1} \right]$$

$$y_1^{(1)} = 0.995867769$$

Second approximation:

Now use updated value,

$$y_1^{(2)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

$$= 1 + \frac{0.1}{2} \left[0 + \frac{1}{x_1^2} - \frac{y_1^{(1)}}{x_1} \right]$$

$$= 1 + \frac{0.1}{2} \left[\frac{1}{1.1^2} - \frac{0.995867769}{1.1} \right]$$

$$y_1^{(2)} = 0.9960555$$

$$\therefore y_1^{(1)} \approx y_1^{(2)} = 0.9960$$

Step 2: Finding $y(1.2)$

$$x_2 = x_1 + h = 1.1 + 0.1 = 1.2$$

So, we have, $x_1 = 1.1$

$$y_1 = 0.9960$$

$$\therefore f(x_1, y_1) = \frac{1}{x_1^2} - \frac{y_1}{x_1} = \frac{1}{1.1^2} - \frac{0.9960}{1.1}$$

$$\therefore f(x_1, y_1) = -0.079008$$

$$\begin{aligned} \Rightarrow y_2^{(0)} &= y_1 + h f(x_1, y_1) \\ &= 0.9960 + 0.1 \times (-0.079008) \\ &= 0.988099 \end{aligned}$$

first approximation:

$$\begin{aligned} y_2^{(1)} &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(0)})] \\ &= 0.9960 + \frac{0.1}{2} \left[-0.079008 + \frac{1}{x_2^2} - \frac{y_2^{(0)}}{x_2} \right] \\ &= 0.9960 + \frac{0.1}{2} \left[-0.079008 + \frac{1}{1.2^2} - \frac{0.988099}{1.2} \right] \end{aligned}$$

$$y_2^{(1)} = 0.9856010$$

Second Approximation :

$$\begin{aligned} y_2^{(2)} &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(1)})] \\ &= 0.9960 + \frac{0.1}{2} \left[-0.079008 + \frac{1}{x_2^2} - \frac{y_2^{(1)}}{x_2} \right] \\ &= 0.9960 + \frac{0.1}{2} \left[-0.079008 + \frac{1}{1.2^2} - \frac{0.9856010}{1.2} \right] \\ &= 0.985705 \end{aligned}$$

$$\text{So, } y_2^{(1)} \approx y_2^{(2)}$$

We can say $y(1.2) = 0.9857$ approx.
Ans.

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